



COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/17/2026	DATE SUBMITTED:	09/17/2026
NAME:	Melquisedec D Iquin	INSTRUCTOR:	Engr. Jimlord Quejado
ACTIVITY NO. 8	Activity 8 Install, Configure, and Manage Availability Monitoring tools		
1. PROCEDURE OUTPUT			

What procedures did you perform?

```
iquin@iquin-VirtualBox: ~/CPE212_Iquin
Reading state information... Done
The following NEW packages will be installed:
  sshpass
0 upgraded, 1 newly installed, 0 to remove and 234 not upgraded.
Need to get 11.7 kB of archives.
After this operation, 35.8 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu noble/universe amd64 sshpass amd64 1
.09-1 [11.7 kB]
Fetched 11.7 kB in 1s (12.6 kB/s)
Selecting previously unselected package sshpass.
(Reading database ... 176979 files and directories currently installed.)
Preparing to unpack .../sshpass_1.09-1_amd64.deb ...
Unpacking sshpass (1.09-1) ...
Setting up sshpass (1.09-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
iquin@iquin-VirtualBox:~/CPE212_Iquin$ ansible-playbook -i hosts nagios_instal
l.yml

PLAY [ubuntu] *****
**

TASK [Gathering Facts] *****
**
```

```
GNU nano 7.2 nagios_install.yml
---
- hosts: ubuntu
  become: true
  roles:
    - nagios

- hosts: centos
  become: true
  roles:
    - nagios
```

```

iquin@iquin-VirtualBox: ~/CPE212_Iquin
TASK [Gathering Facts] *****
**
ok: [192.168.56.123]
ok: [192.168.56.124]

PLAY [centos] *****
**

TASK [Gathering Facts] *****
**
ok: [192.168.56.125]

PLAY RECAP *****
**
192.168.56.123      : ok=1    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
192.168.56.124      : ok=1    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
192.168.56.125      : ok=1    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
iquin@iquin-VirtualBox:~/CPE212_Iquin$

```

```

GNU nano 7.2      main.yml *
---
- name: Install EPEL (CentOS only)
  ansible.builtin.package:
    name: epel-release
    state: present
  when: ansible_distribution == "CentOS"

- name: Install Nagios (Ubuntu)
  ansible.builtin.apt:
    name:
      - apache2
      - nagios4
    state: present
    update_cache: true
  when: ansible_distribution == "Ubuntu"

- name: Install Nagios (CentOS)
  ansible.builtin.dnf:
    name:
      - httpd

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify

```

```
iquin@iquin-VirtualBox: ~/CPE212_Iquin
iquin@iquin-VirtualBox: ~/CPE212_Iquin x iquin@iquin-VirtualBox: ~/CPE212_Iquin/HO... x v
bash: cd: ...: No such file or directory
iquin@iquin-VirtualBox:~/CPE212_Iquin/roles/nagios$ cd ..
iquin@iquin-VirtualBox:~/CPE212_Iquin/roles$ cd ..
iquin@iquin-VirtualBox:~/CPE212_Iquin$ ansible-playbook -i hosts nagios_install.yml -K
BECOME password:

PLAY [ubuntu] *****
**

TASK [Gathering Facts] *****
**
ok: [192.168.56.123]
ok: [192.168.56.124]

TASK [nagios : Install EPEL (CentOS only)] *****
**
skipping: [192.168.56.124]
skipping: [192.168.56.123]

TASK [nagios : Install Nagios (Ubuntu)] *****
**
ok: [192.168.56.124]
```

```
iquin@iquin-VirtualBox: ~/CPE212_Iquin
iquin@iquin-VirtualBox: ~/CPE212_Iquin x iquin@iquin-VirtualBox: ~/CPE212_Iquin/HO... x v
**
changed: [192.168.56.125]

TASK [nagios : Start and enable Nagios (Ubuntu)] *****
**
skipping: [192.168.56.125] => (item=apache2)
skipping: [192.168.56.125] => (item=nagios4)
skipping: [192.168.56.125]

TASK [nagios : Start and enable Nagios (CentOS)] *****
**
changed: [192.168.56.125] => (item=httpd)
changed: [192.168.56.125] => (item=nagios)

PLAY RECAP *****
**
192.168.56.123      : ok=3    changed=0    unreachable=0    failed=0
  skipped=3    rescued=0    ignored=0
192.168.56.124      : ok=3    changed=0    unreachable=0    failed=0
  skipped=3    rescued=0    ignored=0
192.168.56.125      : ok=4    changed=2    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0

iquin@iquin-VirtualBox:~/CPE212_Iquin$
```

```
iquin@iquin-VirtualBox:~/CPE212_Iquin$ ls
HOA_6.1 HOA_7.1 hosts nagios_install.yml Quiz_7.1 README.md roles
iquin@iquin-VirtualBox:~/CPE212_Iquin$ ls
HOA_6.1 HOA_7.1 hosts nagios_install.yml Quiz_7.1 README.md roles
iquin@iquin-VirtualBox:~/CPE212_Iquin$ cd roles/nagios/tasks/
iquin@iquin-VirtualBox:~/CPE212_Iquin/roles/nagios/tasks$ nano main.yml
iquin@iquin-VirtualBox:~/CPE212_Iquin/roles/nagios/tasks$
```

- I basically create a directory for the nagios roles and made nagios_install.yml for the purpose of calling the roles, i fixed the hosts by inputting their ip with their ansible_users. What I did next is created /roles/nagios/tasks which inside is a main.yml file that installs the nagios and its plugins, that goes for the ubuntu and the centos in which centos couldnt dnf install all so what i did was install them one by one.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

- What are the benefits of having an availability monitoring tool?
 - So having this helps detect outages or like degraded services early so often before users can notice them. It reduces the downtime because alerts let admins respond quickly instead of discovering problems very late. It also gives visibility so every server is checked.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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